Validate Community Taxon Count

Description

Validates that a community matrix contains enough taxa for model fitting. Returns the matrix unchanged when the number of taxa is at least 'minimum_taxon_count'. Stops with an informative error when the column count is below the threshold, preventing wasteful model fitting and meaningless species–species associations from near-empty communities.

Usage

```
validate_community_taxon_count(

  data_community_matrix = NULL,

  minimum_taxon_count = 5

)
```

Arguments

`data_community_matrix`	A numeric matrix with samples as rows and taxa as columns, as returned by 'filter_constant_taxa()'.
`minimum_taxon_count`	A single positive integer giving the minimum number of taxa (columns) required to proceed with model fitting. Default is 5.

Details

The check counts 'ncol(data_community_matrix)' after all upstream taxon-level filtering ('filter_community_by_minimum_proportion()', 'filter_community_by_minimum_core_count()', 'filter_community_by_minimum_sample_count()', 'filter_constant_taxa()') has been applied. If the count falls below 'minimum_taxon_count', 'cli::cli_abort()' is called with a message that reports the actual count and the threshold, allowing the user to adjust the configuration or the data.

Value

The input matrix 'data_community_matrix' unchanged, when 'ncol(data_community_matrix) >= minimum_taxon_count'.

See Also

[filter_constant_taxa()], [build_jsdm_fit_input()]